```
!python -m spacy download en_core_web_sm
```

```
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 71.3 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
nlp = spacy.load('en_core_web_sm')
```

```
pip install spacy transformers torch pandas
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
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```

```
import spacy
from spacy import displacy
from transformers import pipeline, AutoTokenizer, AutoModelForTokenClassification
import torch
import pandas
```

```
print("Visualizing Entities with displacy:")
print("-" * 40)

for i, sentence in enumerate(sentences):
    doc = nlp(sentence)
    print(f"\nSentence {i+1}:")
    displacy.render(doc, style="ent", jupyter=True)
    print("-" * 40)
```

```
Visualizing Entities with displacy:
----------------------------------------

Sentence 1:
```

Apple Inc. **ORG** was founded by Steve Jobs **PERSON** , Steve Wozniak **PERSON** , and Ronald Wayne **PERSON** .

```
----------------------------------------

Sentence 2:
```

Google **ORG** 's headquarters are in Mountain View **GPE** , California **GPE** .

```
----------------------------------------

Sentence 3:
```

The Eiffel Tower **LOC** is in Paris **GPE** , France **GPE** , and was designed by Gustave Eiffel **PERSON** .

```
----------------------------------------

Sentence 4:
```

Barack Obama **PERSON** was the 44th **ORDINAL** President of the United States **GPE** .

```
entity_data = []

for sentence_text in sentences:
    doc = nlp(sentence_text)
    if doc.ents:
        for ent in doc.ents:
            entity_data.append({"Sentence": sentence_text, "Entity": ent.text, "Label": ent.labe
    else:
        # If no entities are found, you might want to record the sentence anyway
        # or just skip it. For this task, we'll only record sentences with entities.
        pass
```

```
entity_df = pandas.DataFrame(entity_data)
display(entity_df)
```

	Sentence	Entity	Label
0	Apple Inc. was founded by Steve Jobs, Steve Wo...	Apple Inc.	ORG
1	Apple Inc. was founded by Steve Jobs, Steve Wo...	Steve Jobs	PERSON
2	Apple Inc. was founded by Steve Jobs, Steve Wo...	Steve Wozniak	PERSON
3	Apple Inc. was founded by Steve Jobs, Steve Wo...	Ronald Wayne	PERSON
4	Google's headquarters are in Mountain View, Ca...	Google	ORG
5	Google's headquarters are in Mountain View, Ca...	Mountain View	GPE
6	Google's headquarters are in Mountain View, Ca...	California	GPE
7	The Eiffel Tower is in Paris, France, and was ...	The Eiffel Tower	LOC
8	The Eiffel Tower is in Paris, France, and was ...	Paris	GPE
9	The Eiffel Tower is in Paris, France, and was ...	France	GPE
10	The Eiffel Tower is in Paris, France, and was ...	Gustave Eiffel	PERSON
11	Barack Obama was the 44th President of the Uni...	Barack Obama	PERSON
12	Barack Obama was the 44th President of the Uni...	44th	ORDINAL
13	Barack Obama was the 44th President of the Uni...	the United States	GPE

Feature	spaCy	Hugging Face Tra
Model Type	Rule-based, statistical, CNNs	Transformer-based (BERT, RoBERT
Speed	Generally faster and more lightweight	Can be slower due to larger model
Accuracy (qualitative)	Good for general-purpose NER (e.g., `en_core_web_sm` models)	Often state-of-the-art accuracy du
Context Handling	Local context with some global awareness	Excellent, leveraging deep contextu
Confidence Score	No direct confidence scores for entities (can be derived using custom methods)	Provides direct confidence scores

```
def clean_transformer_output(ner_results):
    cleaned_entities = []
    current_entity = {"word": "", "entity": "", "score": 0.0, "count": 0}

    for item in ner_results:
        word = item['word']
        entity_label = item['entity']
        score = item['score']

        # Check if it's a beginning token (B-) or continuation of a previous entity (I-)
        if entity_label.startswith('B-') or not current_entity["word"] or not entity_label.start
            if current_entity["word"]:
                # Append the previous entity if it exists
                cleaned_entities.append({
                    "Entity": current_entity["word"].replace('##', ''),
                    "Label": current_entity["entity"].replace('B-', '').replace('I-', ''),
                    "Confidence_Score": current_entity["score"] / current_entity["count"]
                })
            # Start a new entity
            current_entity = {
                "word": word,
                "entity": entity_label,
```

```
                    "score": score,
                    "count": 1
                }
            else: # Continuation of an entity (I-)
                current_entity["word"] += word.replace('##', '')
                current_entity["score"] += score
                current_entity["count"] += 1

    # Append the last entity after the loop finishes
    if current_entity["word"]:
        cleaned_entities.append({
            "Entity": current_entity["word"].replace('##', ''),
            "Label": current_entity["entity"].replace('B-', '').replace('I-', ''),
            "Confidence_Score": current_entity["score"] / current_entity["count"]
        })

    return cleaned_entities

all_cleaned_entity_data = []

for sentence_text in sentences:
    ner_results = classifier(sentence_text)
    cleaned = clean_transformer_output(ner_results)
    for entity_info in cleaned:
        all_cleaned_entity_data.append({
            "Sentence": sentence_text,
            "Entity": entity_info['Entity'],
            "Label": entity_info['Label'],
            "Confidence_Score": entity_info['Confidence_Score']
        })

cleaned_transformer_df = pandas.DataFrame(all_cleaned_entity_data)
display(cleaned_transformer_df)
```

	Sentence	Entity	Label	Confidence_Score
0	Apple Inc. was founded by Steve Jobs, Steve Wo...	AppleInc	ORG	0.999405
1	Apple Inc. was founded by Steve Jobs, Steve Wo...	SteveJobs	PER	0.983681
2	Apple Inc. was founded by Steve Jobs, Steve Wo...	SteveWozniak	PER	0.971392
3	Apple Inc. was founded by Steve Jobs, Steve Wo...	RonaldWayne	PER	0.999695
4	Google's headquarters are in Mountain View, Ca...	Google	ORG	0.998801
5	Google's headquarters are in Mountain View, Ca...	MountainView	LOC	0.997765
6	Google's headquarters are in Mountain View, Ca...	California	LOC	0.999266
7	The Eiffel Tower is in Paris, France, and was ...	Eiff	LOC	0.821108
8	The Eiffel Tower is in Paris, France, and was ...	elTower	LOC	0.895773
9	The Eiffel Tower is in Paris, France, and was ...	Paris	LOC	0.999587
10	The Eiffel Tower is in Paris, France, and was ...	France	LOC	0.999647
11	The Eiffel Tower is in Paris, France, and was ...	Gustav	PER	0.998398
12	The Eiffel Tower is in Paris, France, and was ...	eEiffel	PER	0.948596
13	Barack Obama was the 44th President of the Uni...	BarackObama	PER	0.999533
14	Barack Obama was the 44th President of the Uni...	UnitedStates	LOC	0.999509

```
print("Applying Transformer NER to sentences:")
print("-" * 40)

for i, sentence in enumerate(sentences):
    print(f"\nSentence {i+1}: {sentence}")
    ner_results = classifier(sentence)

    if ner_results:
        print("Entities detected (Word | Label | Confidence):")
        for entity in ner_results:
            # Some entities might be broken into sub-words by the tokenizer.
            # We'll display them as they are returned.
            print(f"  - '{entity['word']}' | '{entity['entity']}' | {entity['score']:.4f}")
    else:
        print("  No entities found.")
    print("-" * 40)
```

```
Applying Transformer NER to sentences:
----------------------------------------

Sentence 1: Apple Inc. was founded by Steve Jobs, Steve Wozniak, and Ronald Wayne.
Entities detected (Word | Label | Confidence):
  - 'Apple' | 'B-ORG' | 0.9995
  - 'Inc' | 'I-ORG' | 0.9993
  - 'Steve' | 'B-PER' | 0.9997
  - 'Job' | 'I-PER' | 0.9995
  - '##s' | 'I-PER' | 0.9518
  - 'Steve' | 'B-PER' | 0.9998
  - 'W' | 'I-PER' | 0.9997
  - '##oz' | 'I-PER' | 0.9992
  - '##nia' | 'I-PER' | 0.9993
  - '##k' | 'I-PER' | 0.8590
  - 'Ronald' | 'B-PER' | 0.9997
```

```
  - 'Wayne' | 'I-PER' | 0.9997
----------------------------------------

Sentence 2: Google's headquarters are in Mountain View, California.
Entities detected (Word | Label | Confidence):
  - 'Google' | 'B-ORG' | 0.9988
  - 'Mountain' | 'B-LOC' | 0.9967
  - 'View' | 'I-LOC' | 0.9988
  - 'California' | 'B-LOC' | 0.9993
----------------------------------------

Sentence 3: The Eiffel Tower is in Paris, France, and was designed by Gustave Eiffel.
Entities detected (Word | Label | Confidence):
  - 'E' | 'B-LOC' | 0.9925
  - '##iff' | 'I-LOC' | 0.6498
  - '##el' | 'B-LOC' | 0.7989
  - 'Tower' | 'I-LOC' | 0.9927
  - 'Paris' | 'B-LOC' | 0.9996
  - 'France' | 'B-LOC' | 0.9996
  - 'Gustav' | 'B-PER' | 0.9984
  - '##e' | 'B-PER' | 0.9302
  - 'E' | 'I-PER' | 0.9986
  - '##iff' | 'I-PER' | 0.8858
  - '##el' | 'I-PER' | 0.9797
----------------------------------------

Sentence 4: Barack Obama was the 44th President of the United States.
Entities detected (Word | Label | Confidence):
  - 'Barack' | 'B-PER' | 0.9995
  - 'Obama' | 'I-PER' | 0.9995
  - 'United' | 'B-LOC' | 0.9997
  - 'States' | 'I-LOC' | 0.9993
----------------------------------------
```

```
model_name = "dslim/bert-base-NER"

# 1. Load tokenizer
tokenizer = AutoTokenizer.from_pretrained(model_name)

# 2. Load model
model = AutoModelForTokenClassification.from_pretrained(model_name)

# 3. Create NER pipeline
classifier = pipeline("ner", model=model, tokenizer=tokenizer)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingfa
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or data
  warnings.warn(
config.json: 100%  829/829 [00:00<00:00, 85.5kB/s]
tokenizer_config.json: 100%  59.0/59.0 [00:00<00:00, 8.01kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF H
vocab.txt: 213k/? [00:00<00:00, 15.0MB/s]
added_tokens.json: 100%  2.00/2.00 [00:00<00:00, 216B/s]
special_tokens_map.json: 100%  112/112 [00:00<00:00, 13.4kB/s]
model.safetensors: 100%  433M/433M [00:03<00:00, 206MB/s]
Loading weights: 100%  199/199 [00:00<00:00, 706.90it/s, Materializing param=classifier.weight]
BertForTokenClassification LOAD REPORT from: dslim/bert-base-NER
Key                       | Status     |  |
--------------------------+------------+--+-
bert.pooler.dense.bias    | UNEXPECTED |  |
bert.pooler.dense.weight  | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expe
```

```
from collections import Counter

all_entity_labels = []
for sentence in sentences:
    doc = nlp(sentence)
    for ent in doc.ents:
        all_entity_labels.append(ent.label_)

label_counts = Counter(all_entity_labels)

print("Entity Label Occurrences:")
print("-" * 30)
for label, count in label_counts.items():
    print(f"  - {label}: {count}")
print("-" * 30)
```

```
Entity Label Occurrences:
------------------------------
  - ORG: 2
  - PERSON: 5
  - GPE: 5
  - LOC: 1
  - ORDINAL: 1
------------------------------
```

```
sentences = [
    "Apple Inc. was founded by Steve Jobs, Steve Wozniak, and Ronald Wayne.",
    "Google's headquarters are in Mountain View, California.",
    "The Eiffel Tower is in Paris, France, and was designed by Gustave Eiffel.",
    "Barack Obama was the 44th President of the United States."
]
```

```
print("Applying spaCy NER to sentences:")
print("-" * 40)

for i, sentence in enumerate(sentences):
    doc = nlp(sentence)
    print(f"\nSentence {i+1}: {sentence}")
    print("Entities detected:")
    if doc.ents:
        for ent in doc.ents:
            print(f"  - Text: '{ent.text}', Label: '{ent.label_}'")
    else:
        print("  No entities found.")
    print("-" * 40)
```

```
Applying spaCy NER to sentences:
----------------------------------------

Sentence 1: Apple Inc. was founded by Steve Jobs, Steve Wozniak, and Ronald Wayne.
Entities detected:
  - Text: 'Apple Inc.', Label: 'ORG'
```